

Test On: Not Set

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Abstract

A palindrome reads the same backwards as forwards, ignoring spaces, capitalization, and punctuation. Each word chosen on one side constrains what the other side can complete. We study how sentence patterns can restrict those choices during construction. The search grows two halves outwards and checks whether their observed word tags can still fit a permitted pattern. In one bounded experiment, applying this check during search returned 86,511 distinct mirror pairs, compared with 20,989 when it ran only after closure. The additional pairs span 30 mirror-junction families, compared with 20, so the gain in pair count is larger than the gain in this measure of variety. We describe how pairs can be composed while preserving reversal and separately verify a 90,937-letter dictionary-based construction. The measured outcomes are candidate supply and exact output properties. Readable palindromic prose remains the reason for imposing these constraints, and requires its own evaluation.

1 Introduction

A palindrome is a word or phrase that reads the same backwards as forwards. “Racecar,” “taco cat,” and “step on no pets” all have this property once spaces and capitalization are removed. Checking a palindrome is straightforward: compare its first and last letters, move one position inwards on each side, and continue until they meet.

“Madam, I’m Adam” introduces Adam; “Madam I, madam” says the speaker is mad, using inverted word order (Feldman, 1996, Chapter 3). Both reduce to palindromic *madamimadam*, with different word boundaries and punctuation. We keep word sequences alongside their letters and impose vocabulary, repetition, and sentence-pattern restrictions.

These restrictions can be checked after a candidate is finished. They can also rule out a partial

construction before more time is spent extending it. We compare these two placements of the same tag-pattern test in an outward-growing search. The comparison measures the distinct accepted pairs reached under common stopping limits. A separate dictionary search supplies a long artifact under phrase and token restrictions. The two experiments use different inventories and answer different questions.

2 Building from the middle

Let $n(x)$ lowercase a text and retain only ASCII letters a–z. A text is palindromic when $n(x) = \text{rev}(n(x))$. A mirror pair (L, R) consists of two word sequences such that $n(L) = \text{rev}(n(R))$.

A partial construction stores both word sequences, an unmatched run of letters, and the side that owes that run. We call the unmatched run the *overhang*. The remainder construction follows the approach described by Norvig (2016). Our pair search grows outwards: it prepends words to L and appends words to R . The following trace shows the direction of growth.

Left	Right	What happens next
on		Right owes no
on	no	The overhang is empty
step on	no	Right owes pets
step on	no pets	The overhang is empty

An addition may consume only part of an overhang or extend beyond it. When it extends beyond, the remaining letters become an obligation on the other side. A trie supplies words compatible with the overhang. The trace illustrates this letter operation; its two-word halves do not satisfy the experiment’s three-word minimum.

The search keeps full word sequences because two paths with the same overhang can have different legal continuations under the other restrictions. An empty overhang produces a closed can-

didate. We accept it only if the letter midpoint falls between words, each half contains at least three words, no word occurs twice anywhere in the pair, and both halves pass the complete pattern test. An accepted output is identified by its ordered pair of space-joined halves. Different segmentations remain different outputs, even when the letters coincide.

3 Keeping a sentence pattern possible

For every lowercased word w in the Brown Corpus, we retain its observed tags under NLTK’s universal mapping (Francis and Kučera, 1979; Bird et al., 2009). A word may have several tags. For example, the frozen table allows *credits* as a noun or a verb. We form sentence patterns by removing punctuation and X tags from corpus sentences and keeping patterns of three to nine tags. The frozen payload contains 8,982 distinct patterns.

We keep the 5,649 patterns that contain a VERB and begin with PRON, DET, NOUN, ADJ, NUM, or ADV. Let this set be P and let $T(w)$ be the observed tags of w . A completed sequence $W = (w_1, \dots, w_m)$ passes when

$$F(W) \iff \exists p \in P : |p| = m \wedge \bigwedge_{i=1}^m p_i \in T(w_i). \quad (1)$$

The assignment must fit a whole pattern. Finding a verb somewhere in the tag sets is insufficient on its own.

During construction, the current left words must be able to end a pattern, while the current right words must be able to begin one. Write these tests as $\text{Suf}(L)$ and $\text{Pre}(R)$. The partial-state gate is

$$G(L, R) = \text{Suf}(L) \wedge \text{Pre}(R). \quad (2)$$

Each test allows any observed tag assignment matching the corresponding end of a pattern in P . Empty halves pass; halves longer than nine words fail.

The growth direction explains why this rejection is safe relative to P . A completed left half has the form UL , so any pattern it realizes must have a suffix compatible with L . If no such suffix exists, prepending U cannot make one exist. A completed right half has the form RV , giving the same argument for prefixes. Rejecting a state that fails G therefore removes no completion satisfying F on both halves. A finite stopping limit can

Measure	Terminal	Incremental
Popped states	84,815,872	115,236,212
Closed states	1,117,411	536,805
Partial-state rejections	0	138,210,789
Distinct pairs	20,989	86,511
Process-seconds	19,214.130	19,050.983

Table 1: One recorded search comparison. Process-seconds sum elapsed time across processes. Rejected children do not enter the popped-state count.

still change which of the surviving completions the search reaches.

Passing this test means that a tag assignment fits. Consider this saved accepted pair:

a aaron ama credits erased /
des ares tide rca manor aaa

One admitted reading uses *credits* as a verb. The example also shows why pattern admission needs to be distinguished from grammaticality and meaning.

4 The search comparison

Both configurations use a depth-first stack, compatible word additions, and the same complete acceptance test. Terminal filtering checks patterns after closure. Incremental filtering also checks G on openings and children before they enter the stack. Word uniqueness is checked at closure in both configurations.

The vocabulary file contains 28,402 entries, of which 21,073 occur in the frozen Brown table. Both configurations use this same subset. Outputs contain 20–44 letters; intermediate states have at most 18 words and an overhang of at most 16 letters. The launcher requests 32 CPU processes on one Polaris node, using Python 3.11. Opening i belongs to process $i \bmod 32$, and process ranks 0–31 supply the shuffle seeds.

Within each process, terminal filtering runs first. Each configuration stops at exhaustion or the first of three limits: 600 elapsed seconds, 20 million popped states, or 10,000 distinct accepted pairs. The clock is checked every 4,096 popped states. Results are deduplicated within each process and again across processes. The seeds give the same initial opening shuffle, but pruning changes subsequent random draws and traversal.

Incremental filtering returned 4.12 times as many distinct accepted pairs (Table 1). It also popped more states. Children rejected before insertion are excluded from this counter; their de-

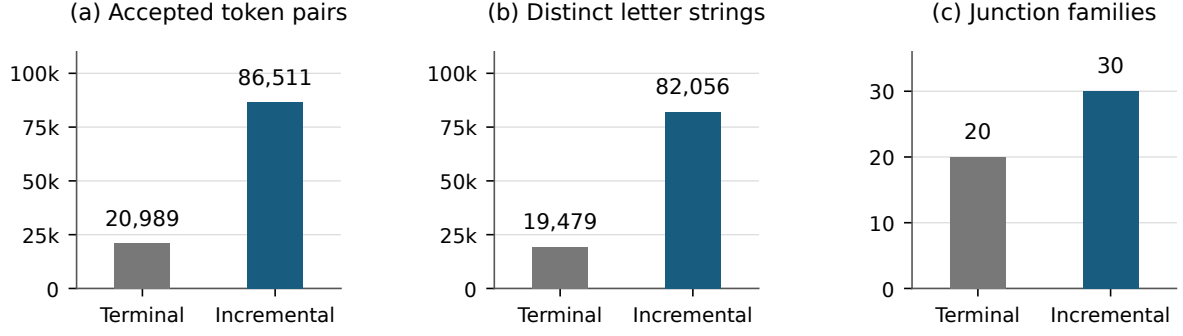


Figure 1: Saved-output counts from the single structural-search comparison. Panels use separate vertical scales. The gain in token pairs and distinct letter strings exceeds the gain in junction families; families are descriptive groups, not independent samples. All values are recomputed from the saved pair lists.

Saved-output measure	Terminal	Incremental
Normalized strings	19,479	82,056
Junction families	20	30
At the 44-letter limit	83.45%	73.49%

Table 2: Additional counts over the same saved pairs. A junction family groups pairs by their two middle words.

scendants are never explored. The larger count describes the states reached after those branches were removed. It does not measure the work of generating children or checking their patterns. Summed elapsed time likewise does not measure CPU time.

We also counted normalized strings and junction families in the saved outputs (Table 2; Figure 1). A junction family is defined by the last word of L and the first word of R . Normalized-string counts show that the yield difference persists after removing alternative segmentations. Family counts grow from 20 to 30, and the incremental family *rates / set* alone contains 10,000 pairs. The gain therefore includes many variations around shared middle words. These families describe concentration; they are not independent samples or a measure of meaning. The configurations share 16,766 accepted token pairs.

The comparison establishes higher output yield in this run. Per-process summaries are absent, so individual stopping causes and workload variation cannot be recovered. Repeated trials with varied seeds, balanced arm order, and recorded stopping reasons are needed to estimate the stability of the difference.

5 Composing pairs and extending length

Once mirror pairs exist, they can be nested around a palindromic center C :

$$L_1 L_2 \cdots L_k C R_k \cdots R_2 R_1. \quad (3)$$

Reversing the letters returns the same sequence. One nesting adjacency creates two surface joins, one on each side; together they form a nesting seam. Our selector makes a seeded greedy pass through a finite bank, accepting pairs that fit a letter ceiling and component repetition limits. Each component receives its own sentence boundary. This gives the pairs a construction procedure, although the search comparison does not evaluate its output quality. Appendix B specifies the caps.

We separately adapted Norvig’s version-3 dictionary search to record normalized letters and maintain changing estimates of the unused phrase inventory. Phrase reuse is forbidden, adjacent repeated tokens are excluded, and tokens outside a specified function-word set may occur at most three times. The saved result contains 90,937 letters and 16,168 distinct normalized phrases. An artifact audit checks the full reversal, dictionary membership, phrase uniqueness, and token restrictions.

Under the same parsing, Norvig’s published reference has 90,439 letters, 16,111 distinct phrases, nine adjacent repeated token pairs, and a maximum nonexception-token count of 14. Our saved output adds 498 letters with the stated restrictions. The objective and feasible set differ from the original search, so this historical reference does not provide a matched-runtime comparison. Available

inventory-policy runs and the exact token exceptions appear in Appendix C.

6 Related work

Norvig (2016) describes remainder-based word placement and a later paired-letter search. These provide the construction mechanisms used here. Papadopoulos et al. (2015) generates palindromes through conjunction of forward and backward corpus graphs. Their Sections 2.1 and 3.4 incorporate syntactic patterns, and their graph construction prunes incompatible states.

Our experiment examines the placement of a finite tag-pattern test in outward-growing overhang search, with separate patterns for the two halves and a terminal word-uniqueness condition. It compares early and terminal checking in that implementation. We do not introduce the general idea of enforcing syntax during palindrome generation or establish an advantage over the earlier graph method.

7 Conclusion

The overhang determines compatible word additions. Sentence patterns rule out partial constructions that cannot pass the completed test. In the recorded search, early checking increased the accepted pair count from 20,989 to 86,511, while junction families increased from 20 to 30. Repeated trials and a common composition procedure can test the value of those additional pairs.

Limitations

The frozen Brown patterns define a finite tag language. They permit ambiguous readings and do not test agreement, argument structure, or a recoverable proposition. The study provides no human evaluation of readability, and composition may run out of acceptable material below its letter ceiling.

The structural comparison has one recorded run per configuration, fixed arm order, and missing process-level traces. Pruning changes the traversal and random-number consumption. The saved aggregate supports exact output counts, but neither a confidence interval for the yield multiplier nor a general runtime claim. The length experiments change both the objective and the feasible set and do not isolate all inventory-policy effects. A longer verified artifact does not establish optimality or a global length record.

Current-file audits confirm properties of the saved outputs. They cannot recover the historical source revision or complete runtime environment, which were not frozen with the structural aggregate. The source and evidence archives require the repository and external inputs to run the full checks.

References

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- Peter Norvig. 2016. [World's longest palindrome? 21,012 words](#). Version 3 (2016); page first created in 2002. Accessed 7 September 2026.
- Alexandre Papadopoulos, Pierre Roy, Jean-Charles Régin, and François Pachet. 2015. [Generating all possible palindromes from ngram corpora](#). In *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 2015)*, pages 2489–2495, Buenos Aires, Argentina.

A Reproducing the saved-output checks

The structural inputs are the frozen files `tools/polaris/payload/brown.json.gz` and `tools/polaris/payload/vocab30k.txt`. The Brown payload contains 49,815 word-tag entries; its sentence-pattern preprocessing removes punctuation and X tags only when forming shapes. Unknown words have no permitted reading. The search planner tests all observed tag combinations without a reading-count cutoff.

The saved terminal and incremental pairs are in `runs/polaris/sentence_plan_20260904_204815/aggregate.json`. The experiment driver and launcher are `tools/polaris/sentence_plan_debug.py` and `tools/polaris/sentence_plan_debug.pbs`. The driver filters the vocabulary before constructing the tries, applies the same complete test in both configurations, and deduplicates ordered token pairs. The launcher combines process outputs and deduplicates them again. Its aggregate does not retain the original per-process traces.

From the repository root, run:

```
cd paper
python3 verify_structural_draft.py
python3 critique_evidence.py
```

The first program checks all 107,500 stored pair rows and the long artifact, and emits hashes of the current inputs and implementation files. The second uses a separate tag checker based on bitsets over pattern positions, without importing the production `SentencePlan` class. It checks vocabulary membership, token uniqueness, complete pattern admission, and reversal, then computes string counts, junction families, lengths, and overlap. Its saved report is `paper/critique-evidence.json`. Both programs belong to this project; neither is an external audit or a rerun of the original timed search.

The input SHA-256 digests are:

```
Brown payload:
14a81b69751dbb9e54251071a97b84fe
3aef9b5532a308028810103acb478419
Vocabulary payload:
6e24f122e2857271de834900c708c7ad
aab806923b3e45f6e363c4bedbcbdb5
Structural aggregate:
ed584869a2fb54cf7bdc7544b8d63a98
b45f87b6b786ef5219f2ca44f03cd22e
```

B Composition settings

The composition selector in `llm_palindrome/hierarchy.py` takes mirror pairs, a center,

and an inclusive letter ceiling. Its counts begin with the center. It scans the supplied pair order once and skips any addition that exceeds the ceiling or the accumulated caps. There is no backtracking. The caller in `server/v3.py` supplies the seeded bank order.

The default limits are two occurrences of an internal bigram, two occurrences of a word-length template, and three occurrences of a token outside the exception set. A word-length template is the sequence of lengths in one component: for example, *rats live* has template (4, 4). Internal bigrams include those within the center and each half; they exclude the joins between components. Added halves may not contain adjacent duplicate words. The procedure counts the center's existing features but does not repair an invalid center.

The composition token exceptions are {a, an, the, of, to, in, on, at, as, is, it, i, for, and, or, if, no, not, was, are, be, by, my, we, me, its, this, that, with, from}. They differ from the exceptions used in the dictionary-length experiment. These are implementation-defined counting sets, not a linguistic classification of all function words.

The bank used by the application applies additional admission and deduplication rules before selection. Consequently, the structural-search pair counts cannot be substituted directly for the size of a composition bank. Evaluating the extra pairs requires passing both configurations through the same bank-construction and selection rules.

C Dictionary-length comparison

The reference is Norvig's published version-3 text (Norvig, 2016). The audit extracts the palindrome body from its HTML, separates phrases at commas, and verifies their normalized forms against the source dictionary. Both texts are lower-cased and word-tokenized as ASCII runs matching `[a-z]+`; punctuation can therefore separate fragments into tokens. The token exceptions for this comparison are {a, an, the, and, or, of, to, in, on, at, for, with, by}.

The saved output, phrase list, result record, and original audit are in `artifacts/norvig-v3/`. The checker is `experiments/audit_norvig_result.py`. All 16,168 output phrases are distinct and occur in the source dictionary. The output has no adjacent repeated words and a maximum of three occurrences of any nonexception token. The reference has 16,111 distinct phrases; its

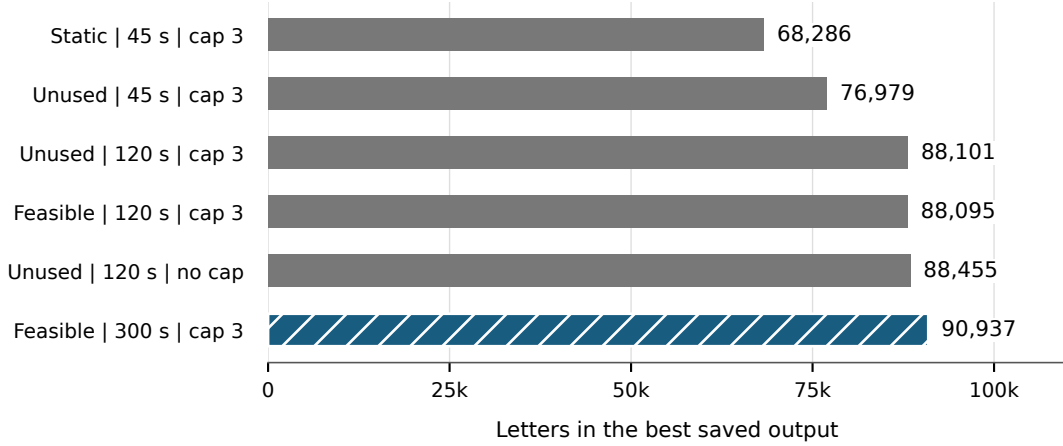


Figure 2: Best saved output from each recorded inventory-policy run. Labels specify the time budget and nonexception-token cap. The horizontal axis begins at zero. These are individual observations with different budgets and constraints, not a scaling curve or a complete ablation.

corresponding token maximum is 14.

Norvig ranks candidate letters using counts of compatible phrase prefixes and suffixes. The adapted policies maintain these estimates differently. Static estimates remain fixed. Unused estimates remove a completed phrase and restore it on undo. Feasible estimates also remove phrases that violate the current used-phrase, character, internal-adjacency, or token-cap conditions. Boundary adjacency is checked when a phrase is admitted.

Inventory	Token cap	Seconds	Letters
Static	3	45	68,286
Unused	3	45	76,979
Unused	3	120	88,101
Feasible	3	120	88,095
Unused	None	120	88,455
Feasible	3	300	90,937

Table 3: Existing runs under the adapted letter objective. All retain phrase uniqueness and the adjacency restriction. These cells do not form a complete ablation.

Figure 2 plots these same observations. The two 45-second runs executed concurrently on the same machine. The other cells have different budgets or token restrictions, and each is a single run. At 120 seconds, unused and feasible estimates differ by six letters. The 300-second endpoint cannot isolate an inventory-policy effect from the longer budget. No matched original-objective arm is available.